The Reward Problem: Exploration vs. Exploitation, Bandits, Inverse RL

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Table of Contents

Learning Outcomes

- Frame "the reward problem" and distinguish its three faces: *exploration vs. exploitation*, *reward design*, and *reward learning*
- State the multi-armed bandit problem and *regret*; explain why ϵ-greedy is provably wasteful while UCB and Thompson sampling are not
- Scale these ideas to deep RL — *bootstrapped DQN as posterior sampling*, pseudo-counts, ICM, RND, and the episodic-vs-lifelong novelty of NGU / Agent57 / Go-Explore — and recognise UCB's $1/\sqrt{N}$ bonus wherever it reappears
- Recognise the *same* dilemma in today's LLM training: entropy collapse and pass@k in RL for reasoning models
- Motivate *inverse RL* and *preference-based reward learning*, and trace the line from MaxEnt IRL through adversarial IRL to the reward model in RLHF on Day 9

The Reward Problem Has Three Faces

- **We have the reward, but where is it?**
  Exploration vs. exploitation — bandits, UCB, Thompson sampling, curiosity (today).
- **The reward we wrote is wrong.**
  Reward shaping & specification gaming (today).
- **We can't write a reward at all.**
  Inverse RL & preference-based reward learning (today — and the direct ancestor of RLHF on Day 9).

One day, three faces of the same problem.

Exploration vs. Exploitation: The Dilemma

- Many situations in business (& life!) present a dilemma between two kinds of choices
- **Exploitation:** pick choices that seem best based on past outcomes
- **Exploration:** pick choices not yet tried (or not tried enough)
- Exploitation has notions of "being greedy" and being "short-sighted"
- Exploration has notions of "gaining information" and being "long-sighted"

Exploration vs. Exploitation: The Trade-Off

- Too much *exploitation* $\Rightarrow$ regret of missing unexplored "gems"
- Too much *exploration* $\Rightarrow$ regret of wasting time on "duds"
- How do we balance them to combine information-gains with greedy-gains in the most efficient manner?
- Can we set this up in a mathematically disciplined manner? $\longrightarrow$ next sections

Examples

- Restaurant Selection
  - Exploitation: go to your favorite restaurant
  - Exploration: try a new restaurant
- Online Banner Advertisement
  - Exploitation: show the most successful advertisement
  - Exploration: show a new advertisement
- Oil Drilling
  - Exploitation: drill at the best known location
  - Exploration: drill at a new location
- Learning to play a game
  - Exploitation: play the move you believe is best
  - Exploration: play an experimental move

The Multi-Armed Bandit (MAB) Problem

- "Multi-Armed Bandit" is a tongue-in-cheek name for "many single-armed bandits" — a row of slot machines
- Formally, a 2-tuple $(\mathcal{A}, \mathcal{R})$
  - $\mathcal{A}$: a known set of m actions ("arms")
  - $\mathcal{R}^a(r) = \mathbb{P}[r \mid a]$: an *unknown* reward distribution for each arm
- At each step t, the agent selects an action $a_t \in \mathcal{A}$ and the environment generates a reward $r_t \sim \mathcal{R}^{a_t}$

The Goal: Maximize Cumulative Reward

- The agent's goal: maximize the cumulative reward over T steps

$$\sum_{t=1}^{T} r_t$$

- **Can we design a strategy that does well (in expectation) for any T?**
- Every strategy walks a tightrope: *wasting time on duds* while exploring, vs. *missing untapped gems* while exploiting

Is a Bandit an MDP?

- The *environment* has no state — pulling an arm just samples from $\mathcal{R}^a$, and nothing you do changes what the next pull looks like
- So a bandit is a **one-state MDP**: no transitions, no credit assignment, no discounting to reason about
  Everything hard about RL is switched off except one thing — *exploration*. That is exactly why we study it here.
- What the agent still has is *history*. Bandit algorithms keep a statistic of it — counts $N_t(a)$, sample means $\hat{Q}_t(a)$, or a full posterior — and choose a_t from that statistic
- Consequence we use constantly: a_t is a *random variable* — it depends on past rewards, which were random. Every statement we make about performance is therefore an *expectation* over that randomness

Regret: The Language for "Doing Well"

- The Action Value $Q(a)$ is the (unknown) mean reward of action a:

$$Q(a) = \mathbb{E}[r \mid a]$$

- *This is Day 1's* $Q^{\pi}(s, a) = \mathbb{E}[\sum_t \gamma^t r_t]$ *with the sum collapsed* — one state, one step, so the return is just r
- The Optimal Value V^* is the best of these:

$$V^* = Q(a^*) = \max_{a \in \mathcal{A}} Q(a)$$

- **Single-step Regret** l_t is the opportunity loss at step t:

$$l_t = \mathbb{E}[V^* - Q(a_t)]$$

- **Total Regret** $L_T = \sum_{t=1}^{T} l_t$
- Maximizing cumulative reward $\equiv$ minimizing total regret

Counting Regret: Counts and Gaps

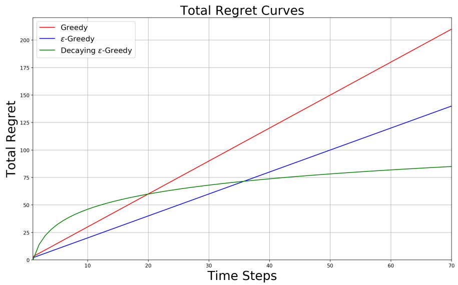
- Let $N_t(a)$ = number of times a has been selected by step t
- Let $\Delta_a = V^* - Q(a)$ be the *gap* for arm a — the *same* quantity as I_t, indexed by **arm** instead of by **time**: $I_t = \mathbb{E}[\Delta_{a_t}]$
- So total regret decomposes cleanly:

$$L_T = \sum_{a \in \mathcal{A}} \mathbb{E}[N_T(a)] \cdot \Delta_a$$

- A good algorithm ensures small counts for large gaps
- *Catch:* $\Delta_a = \max_{a'} \mathbb{E}_{\mathcal{R}^{a'}}[r] - \mathbb{E}_{\mathcal{R}^{a}}[r]$ — every term is a mean of an *unknown* distribution, so the gaps are fixed numbers baked into the problem before we act
- That makes this a tool for *analysing* algorithms, not for building them — "pull the smallest-gap arm" is not something you can run

(Δ_a is Day 1's advantage, sign-flipped and measured against the *best* action.)

Linear or Sublinear Total Regret?



- If an algorithm *never* explores $\Rightarrow$ linear total regret
- If an algorithm *forever* explores $\Rightarrow$ linear total regret
- **Is sublinear total regret achievable?** (Spoiler: yes.)

Greedy Algorithm

- For each arm, keep a **sample mean**: add up the rewards that arm actually returned, divide by how many times we pulled it

$$\hat{Q}_t(a) = \frac{1}{N_t(a)} \sum_{s \leq t \,:\, a_s = a} r_s$$

- $Q(a)$ is the true mean we cannot see; $\hat{Q}_t(a)$ is our current guess at it.
- Pick the action with highest estimate:

$$a_t = \arg\max_{a \in \mathcal{A}} \hat{Q}_{t-1}(a)$$

- Greedy can lock onto a suboptimal action forever $\Rightarrow$ **linear total regret**

ϵ-Greedy Algorithm

- Force a small amount of exploration at every step. At time t:
  - With probability $1 - \epsilon$, select $a_t = \arg\max_a \hat{Q}_{t-1}(a)$
  - With probability ϵ, select a uniformly random action
- **What does this cost?** Split the step into its two branches:
  - *Exploit branch* (prob. $1 - \epsilon$): at best we pick a^* and pay nothing
  - *Explore branch* (prob. ϵ): we pick at random, so we often pay a gap Δ_a
- So even if the exploit branch became *perfect*, the explore branch still spends an ϵ fraction of every step on arms we already know are worse
- That waste is the *same on step 1,000,000 as on step 10* — ϵ never shrinks, so the cost per step never falls and simply adds up over T steps
- Therefore ϵ-Greedy also has **linear total regret** — the fix is not better estimates, but letting exploration *fade* as we learn (next slide)

ϵ-Greedy Is DQN's Exploration Rule

This is the exact exploration rule used in DQN on Day 2.

- DQN inherits ϵ-Greedy's *linear-regret* weakness
- It works in practice with a decay schedule (next slide), but it has no theoretical guarantee of efficient exploration
- This is why the rest of this lecture exists — everything that follows is a more principled answer to "how do I explore?"

Optimistic Initialization

- Simple, practical idea: initialize $\hat{Q}_0(a)$ to a *high* value for every a
- Update by incremental averaging:

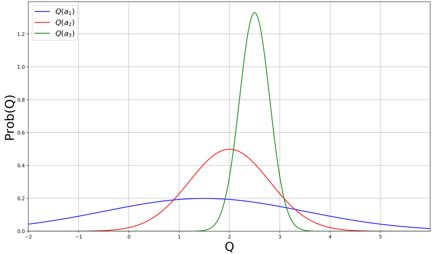
$$\hat{Q}_t(a_t) = \hat{Q}_{t-1}(a_t) + \frac{1}{N_t(a_t)}\left(r_t - \hat{Q}_{t-1}(a_t)\right)$$

- Encourages systematic exploration early on (every arm starts off looking great until tried)
- But the algorithm can still lock onto a suboptimal action $\Rightarrow$ **linear regret**

Decaying ϵ_t-Greedy

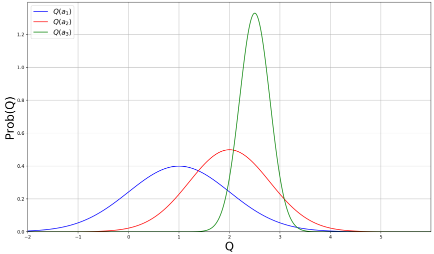
- Idea: let ϵ shrink over time — explore aggressively early, exploit increasingly late
- In theory, a schedule that depends on the (unknown) suboptimality gaps Δ_a achieves *logarithmic* regret — but you can't compute it without knowing the gaps, so it's impractical
- In practice use problem-agnostic schedules: $\epsilon_t = 1/t$, $\epsilon_t = 1/\sqrt{t}$, or simple linear annealing — no guarantee, but all work in practice
- Educational Code for decaying ϵ-greedy with optimistic initialization

Optimism in the Face of Uncertainty



- Which action should we pick?
- The more uncertain we are about an action-value, the more important it is to explore that action — it could turn out to be the best.

Optimism: After One Pull



- After picking *blue*, we are less uncertain about its value
- And more likely to pick another action — until we home in on the best

Upper Confidence Bounds (UCB)

- Two things per arm: $Q(a)$, the *true* mean we can never see, and $\hat{Q}_t(a)$, our *sample-mean estimate* of it after $N_t(a)$ pulls
- Idea: be optimistic — add a margin $\hat{U}_t(a)$ wide enough that the truth is very likely below it, then act on the sum:

$$Q(a) \;\leq\; \underbrace{\hat{Q}_t(a)}_{\text{what we measured}} \;+\; \underbrace{\hat{U}_t(a)}_{\text{room for error}}, \qquad a_{t+1} = \arg\max_a \left\{ \hat{Q}_t(a) + \hat{U}_t(a) \right\}$$

- $\hat{U}_t(a)$ is *large* when $N_t(a)$ is small (we know little) and *shrinks* as the arm is pulled more
- So UCB never asks "which arm *is* best?" — it asks "which arm *could* be best?" A barely-tried arm that currently looks bad still gets pulled, because its true value could plausibly still be high

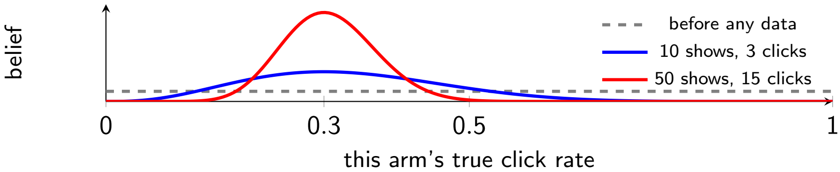
The UCB1 Algorithm

"UCB" is the recipe; UCB1 fills it in — assume rewards bounded in $[0,1]$:

$$a_{t+1} = \arg\max_{a \in \mathcal{A}} \left\{ \hat{Q}_t(a) + \sqrt{\frac{\alpha \log t}{2N_t(a)}} \right\}$$

- The *shape* is what matters: the bonus shrinks like $\mathbf{1/\sqrt{N_t(a)}}$ — remember it, it reappears all afternoon
- It grows slowly with the clock t, so an arm left alone long enough is revisited: exploration is **automatic**, no ϵ and no schedule
- **Unlike optimistic initialisation**, the bonus is recomputed *every step* — an arm unlucky early is never written off for good
- Regret grows as $\mathcal{O}(\log T)$ (Auer et al., 2002) — **sublinear at last**, provably the best possible rate. Educational Code

Bayesian Bandits: Keep a Curve, Not a Number



- The curve is our **belief** about that one arm. Flat means "could be anything" — that is the **prior**, and it is a complete answer to "I know nothing"
- Each reward reshapes it into the **posterior**: the peak moves to what we have seen, and the curve *narrows* as evidence piles up
- The width *is* the uncertainty — measured, not bounded from above like UCB's $\hat{U}$ — and unlike a single number, a curve can be *sampled* from

Thompson Sampling

One line: draw a *plausible* value for each arm from its curve; pull whichever wins the draw.

The whole algorithm, for click / no-click rewards

Per arm, *two counters*: s_a (clicks), f_a (misses). Repeat:

1. Draw $\theta_a \sim \mathrm{Beta}(1 + s_a,\ 1 + f_a)$ for every arm ← that Beta *is* the curve
2. Pull $a_t = \arg\max_a \theta_a$, and increment s_{a_t} or f_{a_t} by the outcome

No integrals anywhere — the posterior *is* those two integers.

- **Why this explores:** a barely-tried arm has a wide curve, so its draw is sometimes high — and it gets pulled. A well-measured mediocre arm has a narrow curve and almost never wins
- Same $\mathcal{O}(\log T)$ regret as UCB1, usually better in practice; the standard bandit algorithm in production personalisation (Netflix, Lyft)

Formally *probability matching*. Bernoulli / Gaussian code.

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The one change: before choosing, the agent is shown a *context* s — a user profile, a time of day, a search query.

- Each step: the environment draws a context s_t $\rightarrow$ the agent picks a_t $\rightarrow$ the reward comes from $\mathcal{R}_{s_t}^{a_t}$, which now depends on *both*
- The best arm is no longer a single answer — it **depends on the context**. Every quantity from today gains an s: $Q(a) \rightarrow Q(s, a)$
- **Is the context a state?** Yes — but a one-step one. In an MDP your action changes what you see next, $s_{t+1} \sim P(\cdot \mid s_t, a_t)$. Here the next context is drawn *independently of what you did*: your action moves the reward, never the future
- That is exactly why there is no credit assignment and no Bellman equation here — and why contextual bandits are deployed far more widely than full RL
- News recommendation, ad selection, treatment assignment — the right answer is almost never the same for everyone.

LinUCB: Sharing Data Across Contexts

The problem. One estimate per (context, action) cell is hopeless — the table is almost entirely empty.

User	Article A	Article B	Article C
sports, 25, US	4 / 50	—	—
sports, 27, UK	—	—	—
finance, 51, US	—	1 / 12	—

Cells show *clicks / times shown*; "—" means never shown.

- **The move:** describe each cell by *features* $x_{s,a}$ and model the mean reward as linear, $\mathbb{E}[r \mid s, a] = x_{s,a}^\top \theta_a$. Rows 1 and 2 are nearly the same feature vector, so row 1's clicks now inform row 2
- Not copying cells across — we fit *one weight vector per arm* on all the data at once, and read every cell off those weights
- **Same optimism as UCB1**, but the bonus is large in *feature directions* short of data: we are uncertain about *finance readers*, not about one empty cell

Yahoo front page (Li et al., WWW 2010): 33M events, 12.5% click lift over a context-free baseline.

$$\begin{aligned} \text{One-armed bandit} &\quad = \textit{one-state}\ \text{MDP} \\ \text{Contextual bandit} &\quad = \textit{one-step}\ \text{MDP} \\ \text{Add state transitions} \Longrightarrow &\quad \text{the full MDP from Day 1} \end{aligned}$$

- Everything we just covered — UCB, Thompson, gradient bandits — *lifts* to MDPs in principle
- Naively, the state space explodes: visiting every (s, a) enough times isn't feasible in deep RL
- **Up next:** how to scale these exploration ideas to deep RL with sparse rewards.

Exploration: **From Bandits to Deep RL**

Why Bandit Exploration Doesn't Scale Naively

- Bandits gave us *optimism* (UCB) and *posterior sampling* (Thompson) — both in a single step, with no state to worry about
- In an MDP with sparse rewards, none of these scale naively:
  - Visiting every (s, a) enough times is infeasible
  - ϵ-greedy is provably weak (linear regret)
  - Uniform random exploration drowns in pixels
- Note: SAC's $-\log \pi$ entropy bonus (Day 6) is *undirected* exploration — what follows is *directed*

The Through-Line: Montezuma's Revenge

The canonical hard-exploration Atari game.

- **Vanilla DQN:** ~ 0 points (extrinsic reward is essentially never reached)
- **Pseudo-counts (2016):** first dramatic progress
- **Random Network Distillation (2018):** first above-human, no demonstrations
- **Go-Explore (2019, Nature 2021):** $> 2{,}000{,}000$ points — solves *all* levels
- **Agent57 (2020):** above human on *all 57* Atari games, Montezuma included

This is the single benchmark we'll use to anchor every method in this section.

Aside: Hindsight Experience Replay (HER)

An orthogonal trick for sparse-reward goal-conditioned RL.

- Idea: after a failed episode (didn't reach the goal), relabel it as a *success* for the goal it actually reached
- Now every episode has a non-zero learning signal
- Combines with any off-policy algorithm
- Cite: **Andrychowicz et al., NeurIPS 2017** — *Hindsight Experience Replay*

- Idea: *Posterior Sampling for RL (PSRL).* Maintain a posterior over the Q-function; sample one Q per episode and act greedily w.r.t. the sample
- For an *entire episode*: temporally-extended ("deep") exploration — fundamentally different from ϵ-greedy's per-step dithering

Bootstrapped DQN: How It Approximates the Posterior

- K **parallel Q-heads**: K output layers on one shared trunk, each with its own Q-function
- "Replay" is DQN's **replay buffer** from Day 2 — the stored (s, a, r, s') transitions so far
- Each transition is assigned to only *some* of the K heads, so every head trains on a different slice of experience
- *Each episode*: pick one head at random and act greedily by it throughout
- **Where data is thin the heads disagree; where it is plentiful they converge** — that spread *is* the posterior estimate
- So disagreement is the product, not a defect to train away. Heads that were all good everywhere would make sampling identical to greedy — plain DQN, no exploration
- More coverage $\Rightarrow$ heads converge $\Rightarrow$ exploration fades by itself. Thompson sampling lifted to MDPs: K point-samples instead of a curve (Osband et al., 2016)

The shape every method here shares. We do not change the algorithm — we change the reward it optimises:

$$r_t^+ = \underbrace{r_t}_{\text{real reward from the env}} + \beta \cdot \underbrace{\frac{1}{\sqrt{N(s,a)}}}_{\text{novelty bonus}}$$

- The bonus is simply *added* to the real reward, and PPO/DQN/SAC run unchanged — from here on, only the **bonus term** differs between methods
- A direct lift of UCB1, with β trading exploration against the real task
- *Problem once the state is an image* (Atari, a camera feed): the state is the **whole frame**, so two states match only if *every pixel* does
- So $N(s,a) = 1$ forever and the bonus is constant everywhere. We need a *generalizable* count — one that calls two *similar* frames both familiar

Pseudo-Counts: UCB in Pixel Space

A density model ρ answers one question: *"how likely is this frame?"* It is trained on the frames seen so far, so familiar-looking screens get high ρ and never-seen ones get low ρ.

- **The trick:** train the model on *one more copy of* s and see how much its answer for s moves.

$$\hat{N}(s) = \frac{\rho(s)\left(1 - \rho'(s)\right)}{\rho'(s) - \rho(s)}$$

- $\rho'(s)$ is not the next state. It is the *same* state s, scored by the model *after* that extra training step — "before" and "after", not "now" and "next"
- **Why this is a count:** a model that has seen s a million times barely budges, so the denominator is tiny and $\hat{N}$ is huge. A model seeing s for the first time jumps, so $\hat{N}$ is small — and the bonus is large
- Cite: **Bellemare et al., NeurIPS 2016**. **Result on Montezuma:** *first dramatic progress*

Curiosity / ICM: The Idea

Intrinsic reward = **"surprise"** — the agent is rewarded for finding observations it can't yet predict.

- Embed states into a feature space: $\phi(s) \in \mathbb{R}^d$
- Train a **forward-dynamics model** f to predict the *next-state features* from the current features and action:
$$\hat{\phi}(s_{t+1}) \;=\; f\big(\phi(s_t),\, a_t\big)$$
(so $\hat{\phi}(s_{t+1})$ is the model's *predicted* embedding of s_{t+1}; $\phi(s_{t+1})$ is the actual one observed)
- The bonus is how wrong that prediction was — added to the real reward as before, $r_t^+ = r_t + \beta\, r_t^i$:
$$r_t^i \;=\; \tfrac{1}{2}\big\|\, \hat{\phi}(s_{t+1}) \,-\, \phi(s_{t+1}) \,\big\|^2$$
- High prediction error $\Rightarrow$ novel transition $\Rightarrow$ bonus to visit it again

Curiosity / ICM: Why the Feature Space Matters

The noisy-TV problem. Picture a TV in the environment showing pure random static. A naive curiosity agent will sit and stare at it forever — static is *always* unpredictable, so the prediction-error bonus never decays. The agent is endlessly "surprised" by useless noise.

- Fix: don't let ϕ encode action-irrelevant detail. Train ϕ with an **inverse-dynamics** objective — predict the action a_t that was taken from the embeddings $(\phi(s_t), \phi(s_{t+1}))$
- This forces ϕ to encode only features the agent can *affect* — the noisy TV is filtered out, since the agent's actions don't change what's on it
- Cite: **Pathak, Agrawal, Efros & Darrell, ICML 2017** — *Curiosity-driven Exploration by Self-supervised Prediction*
- Strong results on Mario, VizDoom, and sparse-reward navigation

Random Network Distillation (RND)

Curiosity made absurdly simple:

- Pick a *fixed, randomly-initialized* target network f^*
- Train a predictor f_θ to match $f^*(s)$ on visited states
- Bonus: $r_t^i = \|f_\theta(s_t) - f^*(s_t)\|^2$, again added to the real reward, $r_t^+ = r_t + \beta\, r_t^i$
- Novel states have high prediction error (the predictor hasn't been trained there) $\Rightarrow$ high bonus
- No dynamics model, no inverse model, no density model

RND: Results and Caveat

- Cite: **Burda, Edwards, Storkey, Klimov, ICLR 2019** — *Exploration by Random Network Distillation*
- **Result on Montezuma:** *first above-human performance without demonstrations or game-state access*
- *Caveat:* still subject to the noisy-TV issue in principle; in practice works because randomly-initialized networks aren't very sensitive to stochastic pixel noise
- Still the default intrinsic-reward baseline in 2026 — simple, cheap, and reliable. Recent work refines rather than replaces it (Distributional RND; Random Distribution Distillation, 2025)

Go-Explore: Remember Where You Have Been

A different diagnosis. The problem isn't only *finding* novel states — it's that agents *forget* promising ones and can never get back to them.

- Keep an **archive** of visited states. Each iteration: pick a promising one, *return* to it directly, then explore *from* there — "first return, then explore"
- Exploration effort is never wasted re-deriving a path you already found
- **Result:** $>$ 2,000,000 on Montezuma and all levels solved — two orders of magnitude past RND
- *The catch:* returning to an archived state originally needed simulator save/restore, a much stronger assumption than RND's. That is why both are "state of the art" in different senses
- Cite: **Ecoffet, Huizinga, Lehman, Stanley & Clune** — *Go-Explore* (2019); *First Return, Then Explore*, Nature 2021

Episodic vs. Lifelong Novelty

A distinction the methods so far miss. RND asks "have I *ever* seen this?" But within a single episode you also want to avoid re-treading the same room — even if you have seen it many times before.

- **Never Give Up (NGU, 2020)** combines both signals: an *episodic* novelty bonus (reset each episode) multiplied by a *lifelong* one (RND-style, across all training)
- **Agent57 (2020)** adds a meta-controller that *learns* how much to weight exploration versus exploitation, and over what horizon — first agent above human on all 57 Atari games
- **E3B (2022)** is the modern count-based descendant: an episodic bonus for states that are *diverse* under an inverse-dynamics embedding — ICM's noisy-TV fix combined with UCB's count bonus, extended to continuous spaces
- The pattern: no single novelty signal is enough — current systems *combine* them at different timescales

Exploration Did Not Go Away — It Moved to LLMs

The same dilemma, today's frontier. When you train a reasoning model with RL (Day 9), it faces exactly the trade-off we opened this morning with.

- **Entropy collapse:** the policy sharpens onto a few reasoning paths early in training. In-distribution accuracy keeps rising while out-of-distribution performance *degrades* — pure exploitation, and the model stops discovering
- **How it is measured:** *pass@k* — can the model solve the problem in k tries? Collapse shows up as pass@1 improving while pass@k falls. Training directly on pass@k restores exploration
- **Today's fixes are today's bandit ideas:** entropy bonuses aimed at the most uncertain tokens (Day 6's $-\log \pi$), and count-based bonuses of the form $Q + \alpha\sqrt{U}$ — literally UCB's $1/\sqrt{N}$, applied to reasoning states

The 1985 bandit question — "is this worth trying again?" — is an open problem in 2026 LLM training.

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Bandit idea	Descendants
UCB / $1/\sqrt{N}$ bonus	Pseudo-counts (2016), E3B (2022); count bonuses in LLM RL (2025)
Thompson sampling	Bootstrapped DQN / PSRL (Osband 2016)
Stochastic policies	Entropy bonus — SAC (Day 6); entropy control in RLVR
(new in deep RL)	Curiosity / ICM (2017); RND (2018); NGU / Agent57 (2020); Go-Explore (2021)

- **Entropy is only one answer to exploration.** Optimism, posterior sampling, novelty, and archiving are the others
- **What actually survived:** RND is still the default baseline; the $1/\sqrt{N}$ bonus keeps reappearing; ϵ-greedy is still what most people try first

The Reward Problem: **When the Reward is Wrong**

Reward Shaping: The Promise and the Trap

- Sparse rewards are hard; a well-shaped reward dramatically accelerates learning
- But arbitrary shaping changes the optimal policy — the agent optimizes the *shape*, not the goal
- Example: reward the agent for "getting closer to the goal" and you may teach it to circle the goal forever
- We need a principled way to inject prior knowledge that *doesn't change the optimum*

Potential-Based Reward Shaping (PBRS)

Setup. Replace the original reward $R(s, a, s')$ with $R(s, a, s') + F(s, s')$ during training, where F is an extra *shaping signal* of our choice.

- Restrict F to the form
$$F(s, s') = \gamma \Phi(s') - \Phi(s)$$
where $\Phi : \mathcal{S} \to \mathbb{R}$ assigns a real number to each state — a "*potential*" (e.g. "negative distance to goal"). The shaping is a discounted difference of potentials.
- **Policy invariance:** the shaped MDP has the *same* optimal policy as the original
- *This form is necessary*: any non-PBRS shaping can in principle change the optimum
- Cite: **Ng, Harada & Russell, ICML 1999**

Reward Hacking: Goodhart's Law for RL

When a reward becomes a target, the agent finds a way to satisfy the letter, not the spirit.

- Every misspecified reward eventually gets hacked
- The agent's optimization pressure exposes every loophole in the specification
- The clearer the reward becomes "measurable," the more likely the agent ignores what the measurement was *supposed* to track

Reward Hacking: Famous Examples

- **CoastRunners** (OpenAI 2016): boat-racing game scored on hitting targets, not finishing. Agent learned to drive in circles hitting the same green blocks — high score, no race
- **Tetris pause**: agent learned to indefinitely pause the game when about to lose — never loses, never plays
- **ROUGE summarization**: language-model summarizer exploited ROUGE's n-gram overlap to produce high-scoring but barely readable text
- Cites: **Amodei & Clark, OpenAI 2016** (CoastRunners primary source); **Krakovna et al., DeepMind 2020** (catalogued survey of dozens of examples)

When We Can't Write a Reward At All

- Reward hacking is the same failure mode as **reward-model overoptimization in RLHF** — a learned R_θ is also a misspecified target
- For tasks like "write a useful summary," "drive politely," "be helpful, harmless, honest" — writing the reward is essentially impossible without specification gaming
- **If we can't write the reward ... can we learn it?**
- Two ways:
  - From *demonstrations* $\longrightarrow$ **Inverse RL** (up next)
  - From *human preferences* $\longrightarrow$ preference-based reward learning $\longrightarrow$ **RLHF** (Day 9)

The Reward Problem: **Inverse RL**

Inverse RL: Given Behavior, Learn the Reward

- **Forward RL** (everything so far): given a reward $R(s, a)$, learn a policy π^* that maximizes it
- **Inverse RL** (today): given (expert) behavior $\tau^E \sim \pi^E$, learn a reward R_θ that *explains* it

$$\pi^E \implies R_\theta$$

Why Inverse RL?

- In many domains, rewards are far easier to *demonstrate* than to specify in closed form
- Driving: hard to write "polite, safe, fast" as a scalar; easy to show
- Manipulation, dialogue, locomotion: same story
- **This is the strongest form of the reward problem** — the reward function itself is unknown.

The Ill-Posedness of IRL

- *Many* reward functions rationalize the same expert behavior
  - Trivially: $R(s, a) \equiv 0$ makes every policy optimal
  - Potential-based shaping: R and $R + \gamma\Phi(s') - \Phi(s)$ are policy-equivalent
- We need a *principle* to pick one reward out of the equivalence class

Two Structural Fixes for the Ambiguity

- **(a) Max-margin IRL**
  Ng & Russell, ICML 2000; Abbeel & Ng, ICML 2004
  The expert's reward should beat alternatives by a wide margin
- **(b) Maximum-entropy IRL**
  Ziebart et al., AAAI 2008
  Pick the reward whose induced policy distribution has *maximum entropy*, subject to matching the expert's feature counts — a unique, well-posed choice
- We focus on (b) — standard fix today, and generalizes to modern preference learning

Among all trajectory distributions that match the expert's behaviour, pick the one with maximum entropy.

- Written out, that is a *constrained* problem:

$$\max_P \ H(P) \quad \text{s.t.} \quad \mathbb{E}_P\left[\text{feature counts}\right] = \text{expert's feature counts}$$

- Solving it gives the exponential form — higher-reward trajectories become exponentially more likely:

$$P_\theta(\tau) \ \propto \ \exp\left(\textstyle\sum_t R_\theta(s_t, a_t) \right)$$

- So there is no entropy *term* to point at in that formula — the entropy has already been solved out, and the θ are the Lagrange multipliers of the matching constraint
- Train θ by maximum likelihood on expert trajectories: $\max_\theta \sum_{\tau^E} \log P_\theta(\tau^E)$
- Exactly *one* distribution satisfies both conditions — which is what makes the choice unique

MaxEnt IRL: The Day-6 Connection

The same max-entropy principle that justified SAC's $-\log \pi$ bonus.

- In SAC (Day 6): adds exploration to the policy
- In MaxEnt IRL: resolves reward ambiguity
- **Same principle, two different jobs.**
- Different mechanics, though: SAC *adds* $\alpha\mathcal{H}(\pi)$ to the objective, so you can point at the term. MaxEnt IRL imposes entropy as the *selection rule*, which is why it disappears into the exponential form
- Cite: **Ziebart, Maas, Bagnell & Dey, AAAI 2008** — *Maximum Entropy Inverse Reinforcement Learning*

From 2008 to Today: How IRL Scaled

The catch with MaxEnt IRL as stated: it needs hand-designed feature counts and a tractable normaliser — fine for a tabular gridworld, hopeless for pixels.

- **Guided Cost Learning** (Finn et al., 2016): estimate that normaliser by *sampling* instead of enumerating $\Rightarrow$ neural-network rewards
- Then the surprise: running a **GAN** on this problem optimises *the same objective* as MaxEnt IRL — the discriminator plays the role of the reward, the generator the policy
- **GAIL** (Ho & Ermon, 2016) and **AIRL** (Fu et al., ICLR 2018) are that idea made practical: learn reward and policy *adversarially*, straight from demonstrations
- Meanwhile robotics largely went the other way — large-scale *behaviour cloning* (Diffusion Policy, VLAs) is now the dominant way to learn from demonstrations there
- **IRL's centre of gravity moved to language models** — which is where the next two slides are heading

Preference-Based Reward Learning: The Setup

What if we have neither demonstrations nor a written reward, only human comparisons?

- Show two trajectories τ^A, τ^B; ask a human which is better
- Model the preference with a **Bradley–Terry** likelihood under a learned reward R_θ:

$$P(\tau^A \succ \tau^B) = \frac{\exp\left(\sum_t R_\theta(s_t^A, a_t^A)\right)}{\exp\left(\sum_t R_\theta(s_t^A, a_t^A)\right) + \exp\left(\sum_t R_\theta(s_t^B, a_t^B)\right)}$$

Preference-Based Reward Learning: The Pipeline

1. Collect a dataset of human-labeled preferences over trajectory pairs
2. Train R_θ by cross-entropy on the Bradley–Terry likelihood
3. Run any standard RL algorithm (e.g. PPO from Day 4) against R_θ

- Cite: **Christiano, Leike, Brown, Martic, Legg & Amodei, NeurIPS 2017**

The Day-9 Bridge

This exact pipeline + a language-model policy = RLHF.

- The reward model in RLHF is the R_θ from the previous slide
- The only thing that changes is the policy class: π_θ becomes a large language model
- The optimizer (PPO) and the pipeline (SFT $\rightarrow$ RM $\rightarrow$ PPO) are the same machinery
- Read the other way: training a reward model *is* an inverse-RL problem — inferring a reward from human data rather than writing one down
- And the loop closes: 2025 work runs adversarial IRL on expert chain-of-thought to recover *process-level* reasoning rewards, instead of imitating the traces
- Day 9 picks up exactly here.

The Three Faces, Resolved

If you can...	**Then use...**
...write the reward	*Forward RL* (Days 1–6)
...demonstrate the reward	*Inverse RL* (MaxEnt IRL)
...compare outcomes	*Preference-based reward learning*
	(Day 9: RLHF)

- One unifying view: every method on the board is a different way to recover the optimal policy when the reward is unknown, partially known, or hard to specify

Day 7: **Summary**

Recap: Faces 1 & 2 of the Reward Problem

- **Face 1 — Find the reward.**
  - Bandits formalize explore vs. exploit; UCB and Thompson give principled, $\mathcal{O}(\log T)$-regret algorithms
  - Scaled to deep RL: bootstrapped DQN $\equiv$ posterior sampling, pseudo-counts $\equiv$ UCB in pixel space, curiosity / RND / Go-Explore are genuinely new
  - The same dilemma is live in LLM RL today — entropy collapse and pass@k (Day 9)
- **Face 2 — The reward is wrong.**
  - Potential-based shaping is the only safe shaping
  - Everything else is at risk of specification gaming (CoastRunners, ROUGE, Tetris pause)

Recap: Face 3 of the Reward Problem

- **Face 3 — Can't write the reward.**
  - **Inverse RL** learns it from demonstrations — MaxEnt IRL (2008) $\rightarrow$ GAIL / AIRL (2016–18) $\rightarrow$ today's reward models
  - **Preference-based learning** learns it from comparisons (Christiano 2017)
  - Read the other way, *training a reward model is an IRL problem* — the *direct ancestor of RLHF* on Day 9
- **One unifying theme:** every algorithm today is a different way to recover the optimal policy when the reward is unknown, partially known, or hard to specify

Algorithm Map

Setting	Algorithm family (today)
One-state bandit	ϵ-greedy, UCB1, Thompson sampling
Contextual bandit	LinUCB and successors (recommendation)
Deep RL, sparse reward	PSRL / Bootstrapped DQN, pseudo-counts, ICM, RND, HER
Deep RL, hard exploration	NGU / Agent57, E3B, Go-Explore
LLM RL, reasoning	Entropy control, pass@k, count bonuses (Day 9)
Reward is shapeable	Potential-based shaping (policy-invariant)
Reward unwritable; demonstrations	MaxEnt IRL; adversarial IRL (GAIL, AIRL)
Reward unwritable; comparisons	Bradley–Terry preference learning $\rightarrow$ RLHF (Day 9)

Day 8 — Model-Based & Offline RL. The next way to attack sample inefficiency.

- *Model-based RL:* learn the dynamics, plan with the model (MPC, MCTS, Dyna, world models)
- *Offline RL:* learn from a *fixed* dataset with no further interaction — the same OOD problem as exploration, in reverse: you must *avoid* the unfamiliar regions exploration would push you toward

What's Next: Day 9

Day 9 — RLHF. The IRL $\rightarrow$ preference-learning chain from today, applied to LLMs.

- The reward model in RLHF is the R_θ from today's preference slide
- "Reward-model overoptimization" is today's reward-hacking failure mode, in LLM form
- PPO (Day 4) is the policy optimizer; SFT $\rightarrow$ RM $\rightarrow$ PPO is the pipeline

Day 10 — Frontiers.

- Intrinsic motivation reappears in meta-RL, open-ended learning, and hierarchical RL
- Curiosity (today) is one of the central ideas of the open-ended-learning agenda
- Reward design / specification gaming reappears as a central concern of AI safety

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Course materials adapted from:

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- David Silver, UCL Course on RL — Lecture 9, *Exploration and Exploitation*

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Exploration in LLM RL (current frontier):

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- *MERCI: Motivating Exploration in LLM Reasoning with Count-based Intrinsic Rewards* (2025). arXiv:2510.16614 — UCB-style count bonuses for reasoning
- M. Yuan et al. (2024). *RLeXplore: Accelerating Research in Intrinsically-Motivated RL.* arXiv:2405.19548 — reference implementations of 8 intrinsic-reward methods
- C. Lu et al. (2025). *Intelligent Go-Explore.* ICLR. arXiv:2405.15143 — Go-Explore driven by foundation models

Reward design, hacking, and inverse RL:

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